LineaLinkR

Trajectory-resolved temporal ligand–receptor communication with bifurcation-aware roles

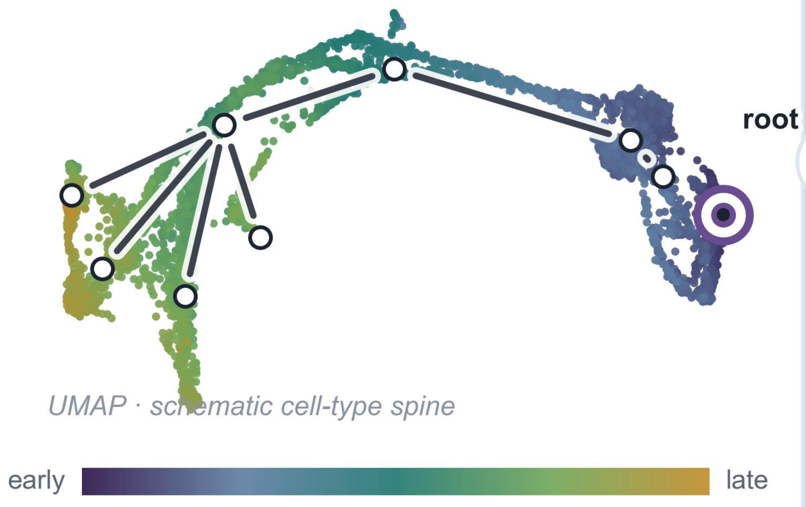
single-cell snapshot → signaling along reconstructed trajectories

01 · INPUT

1

Snapshot state space

counts → PCA / kNN graph



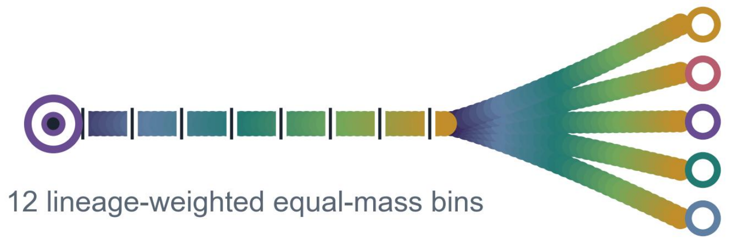
counts + LR prior

02 · TRAJECTORY

2

Lineages & time

geodesic pseudotime → bins



12 lineage-weighted equal-mass bins

- expression-only endpoint-selected attractor root
- Dijkstra geodesic pseudotime
- MST lineages + soft weights → $\psi_{i\ell}$

5 lineages recovered

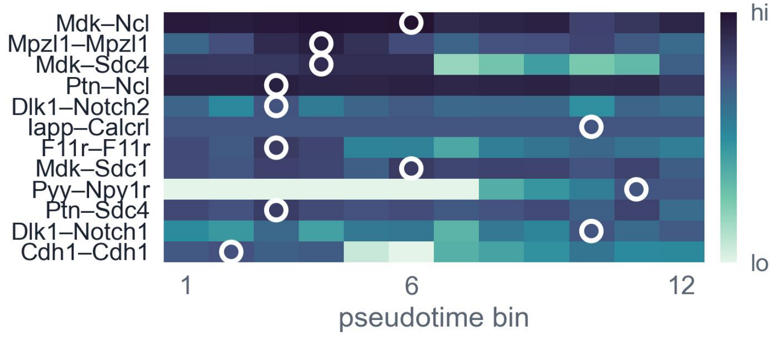
03 · SCORING

3

Temporal LR programs

peak-lineage envelope · display gaps filled

$$A_{LR} = \frac{\sqrt{L \cdot R}}{S + \sqrt{L \cdot R}}$$

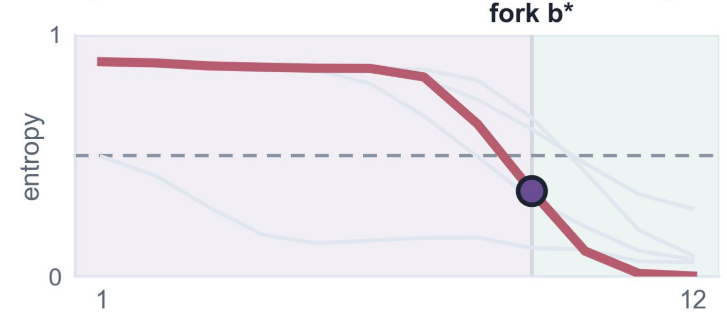


04 · INTERPRET

4

Bifurcation → roles

five operational roles · transition score auxiliary



pre-fork

- ● Instructive
- ● Lateral

post-fork

- ● Reinforcing
- ● Feedback

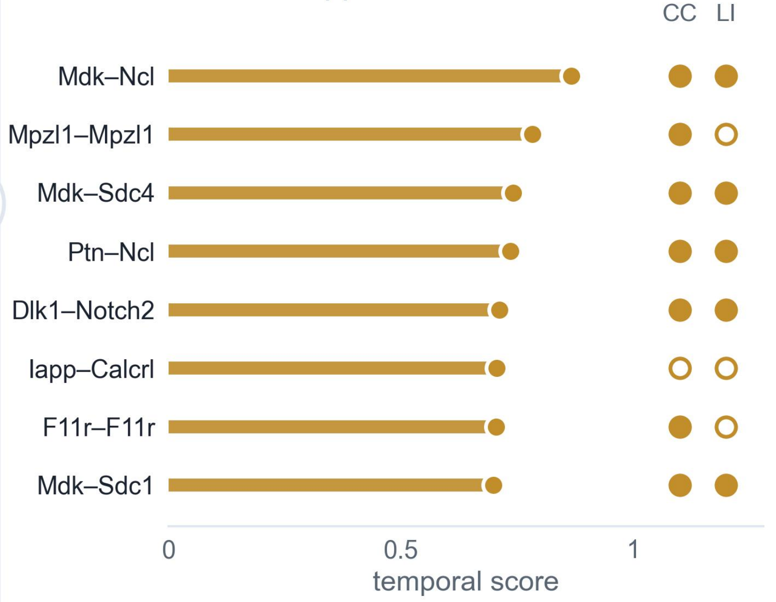
● Bystander

05 · EVIDENCE

5

Ranked programs

score + external support



EVIDENCE real and simulated data · blind ordering · external biological checks

37 broad evaluation

registered dataset slots

24 common-data accuracy

ordering sets

4 terminal F1 + ARI

branch-recovery sets

11 method-specific adapters

eligible TI comparators

5 · 12 timing taxonomy

operational roles · bins